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PULSELIKE CHARACTER OF BLACKBODY RADIATION FROM NEUTRON STARS

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ABSTRACT

The magnetic poles of a neutron star are expected to be hotter than the remainder of its surface. The observed X-ray intensity of blackbody radiation will then be modulated as rotation carries each hot spot across the visible disk of the star. Two models for the distribution of temperature over the stellar surface are presented, and the predicted "light" curves and spectra synthesized.

We then interpret the X-ray emission from two radio pulsars as examples of this effect. Both the light curve and flux from PSR 1509−58 are well described by a model in which the polar caps are strongly heated by the magnetosphere. The light curve of PSR 1055−52, if blackbody in origin, implies that it is nearly aligned, a conclusion of particular interest in that this source shows interpulses.

Subject headings: pulsars — radiation mechanisms — stars: neutron — X-rays: sources

I. INTRODUCTION

X-ray observations with the *Einstein Observatory* (*HEAO 2*) are capable of detecting thermal radiation from neutron stars if their temperatures exceed several times 10^5 K in the best cases (Helfand, Chanan, and Novick 1980). A number of point sources coincident with radio pulsars have been found, some of which may be blackbody in spectrum. Here we analyze a feature of this emission which has not been sufficiently discussed, viz., that it should be pulsed. The magnetic poles of a neutron star are expected to be significantly hotter than the remainder of its surface. The observed X-ray intensity will be modulated as rotation carries each hot spot across the visible disk of the star (Smoluchowski 1972). Because the emission is blackbody in nature, each pole radiates isotropically rather than into a narrow cone: the signature of the process is a broad pulse filling a significant fraction of the pulsar duty cycle, as opposed to the more narrow pulses characteristic of the radio emission.

In § II we roughly analyze the expected distribution of temperature over the surface of a neutron star, and in § III the "light" curve and spectrum of the source are described. Section IV interprets the X-ray emission from the recently discovered pulsar PSR 1509−58 as an example of this phenomenon, and uses the *absence* of modulation in the X-ray emission from PSR 1055−52 to constrain the geometry of the source.

II. DISTRIBUTION OF TEMPERATURE OVER THE SURFACE OF THE STAR

a) Polar Cap Heating

Within a wide variety of models a significant flux of energy is deposited onto the polar cap from the pulsar magnetosphere. Estimates of polar temperatures expected from this process (Cheng and Ruderman 1980; Helfand, Chanan, and Novick 1980; Arons 1981) predict fluxes potentially observable with the *Einstein* observatory and temperatures greater than 10^6 K.

We expect such high temperatures to be confined very closely to the polar cap itself. The thermal conductivity of neutron star matter increases with density, increasing roughly as ρ^2 near the surface in the calculations of Flowers and Itoh (1976: see their Fig. 8). The density itself increases inward extremely rapidly. Very close to the polar cap heat diffuses outward in all directions. However, once it has propagated even a matter of meters, it encounters matter of high thermal conductivity below the surface, which effectively channels its flow downward and limits any further horizontal spread. Furthermore, the transport of energy perpendicular to magnetic field lines, i.e., horizontally at the polar cap, is a far slower process than transport parallel to the field (Canuto and Chiuderi 1970; Itoh 1975). We therefore expect the high-temperature region to extend only slightly beyond the polar cap itself.

The radius of the polar cap is

$$R_{\rm cap} = R^{3/2}(2\pi/cp)^{1/2} \ll R \,, \tag{1}$$

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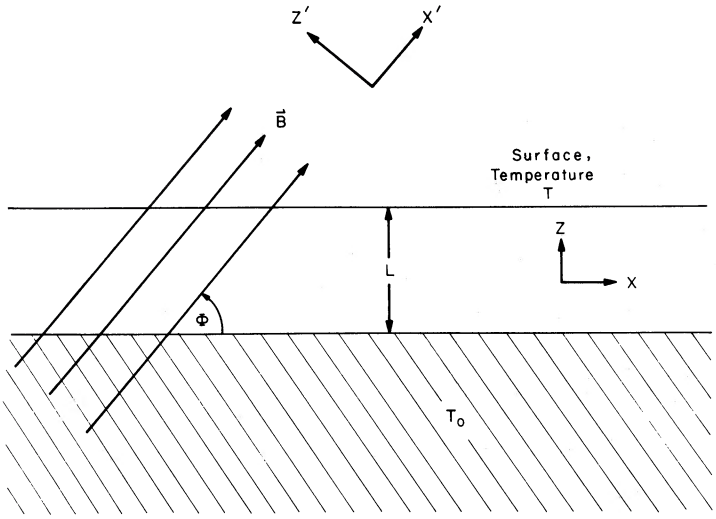

FIG. 1.—Illustrating the geometry of the regime of anisotropic heat conduction

where R is the stellar radius and p the pulsar period. Thermal emission from the cap will dominate that of the rest of the star if

$$\pi R_{\rm cap}^2 T_{\rm cap}^4 > 4\pi R^2 T^4 , \tag{2}$$

or

$$T_{\rm cap}/T > 12 p^{1/4} , \tag{3}$$

where T is the background temperature of the stellar surface and we have assumed $R = 10^6$ cm. This condition is relatively easy to meet: in the case of pulsar PSR 1929+10, for example, Helfand, Chanan, and Novick (1980) estimate $T_{\rm cap} = 4.2 \times 10^6$ K, and we find emission from the cap will dominate if the remainder of the surface lies below 5.1×10^5 K.

b) *Anisotropic Conduction from Below*

Whether or not equation (3) is satisfied, a second process acts to insure nonuniformities in the surface temperature field. The stellar surface lies a matter of meters above the interior, an essentially isothermal reservoir of heat at a far higher temperature. The conduction of heat to the surface from within is channeled by the magnetic field, being far slower across field lines than along them. This phenomenon gives rise to a very slow variation of temperature over the star, the temperature reaching a maximum at the magnetic poles and a minimum along the magnetic equator.

The thickness of the crustal layers in which heat conduction is anisotropic is very small compared to the radius of the star. The conductivity will become isotropic and the temperature constant on horizontal surfaces once a depth is reached at which the one-dimensional electron Fermi energy greatly exceeds the cyclotron energy. This condition obtains for densities

$$\rho \gg 10^4 B_{12}^{3/2} \text{ g cm}^{-3} , \tag{4}$$

where B_{12} is the magnetic field in units of 10^{12} gauss. Alternatively, requiring the Fermi energy to exceed the ionization potential for the last atomic electron (Cohen, Lodenquai, and Ruderman 1970) leads to the condition

$$\rho \gg 1.3 \times 10^3 \text{ g cm}^{-3} . \tag{5}$$

Both equations (4) and (5) are satisfied at very shallow depths.

The heat conduction problem is therefore one-dimensional. Figure 1 illustrates the geometry: a heat reservoir at constant temperature T_0 lies a distance L beneath the surface which is at temperature T ($\ll T_0$). Magnetic field lines thread the slab at an angle Φ to the horizontal. An unprimed coordinate system is oriented such that the x-axis is horizontal, a primed system such that x' points along $\boldsymbol{B}$.

We assume the thermal conductivity tensor to be diagonal in the primed system:

$$\mathbf{K}_{ij} = \begin{pmatrix} K_{\parallel} & 0 \\ 0 & K_{\perp} \end{pmatrix} , \tag{6}$$

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where $K_{\parallel}$ is the conductivity along $\boldsymbol{B}$ and $K_{\perp}$ that perpendicular to it ($K_{\perp} \ll K_{\parallel}$). The transformation of coordinates to the unprimed system is that of a rotation through the angle Φ. Under this rotation K transforms as a tensor:

$$\mathbf{K}_{ij} = \sum_{k,l} \frac{\partial x_i}{\partial x'_k} \frac{\partial x_j}{\partial x'_l} \mathbf{K}'_{kl} , \tag{7}$$

and we obtain the conductivity in the unprimed system:

$$\begin{aligned} K_{11} &= K_{\parallel} \cos^2 \Phi + K_{\perp} \sin^2 \Phi , \\ K_{12} &= K_{21} = (K_{\parallel} - K_{\perp}) \cos \Phi \sin \Phi , \\ K_{22} &= K_{\parallel} \sin^2 \Phi + K_{\perp} \cos^2 \Phi . \end{aligned} \tag{8}$$

The heat conduction equation giving the ith component of the heat flux vector $\boldsymbol{q}$ in the unprimed system is

$$q_i = -\sum_j K_{ij} \partial_j T , \tag{9}$$

from which we obtain the flux upward (with $q_2 = q_z$):

$$q_z = -K_{22} \frac{\partial T}{\partial z} - K_{21} \frac{\partial T}{\partial x} . \tag{10}$$

The boundary condition is that the flux arriving at the surface from below is balanced by the emission of blackbody radiation:

$$q_z \text{ (surface)} = \sigma T^4 . \tag{11}$$

To order of magnitude,

$$\frac{\partial T}{\partial z} \approx \frac{T - T_0}{L} \approx -\frac{T_0}{L} , \qquad \frac{\partial T}{\partial x} \approx -\frac{T}{R} , \tag{12}$$

where R is the stellar radius. Because $R \gg L$ and $K_{22} \approx K_{21}$ to order of magnitude, the $\partial T/\partial x$ term in equation (10) is negligible compared to the $\partial T/\partial z$ term, and we obtain for the surface temperature

$$\sigma T^4 = -(K_{\parallel} \sin^2 \Phi + K_{\perp} \cos^2 \Phi) \frac{\partial T}{\partial z} , \tag{13}$$

where all quantities are evaluated at the surface.

A detailed evaluation of $\partial T/\partial z$ is beyond the scope of this paper. By equation (12) it is to order of magnitude constant, and nothing in the physics of the problem will make it vary rapidly with location. At most it is a weakly varying function of position on the star, and in what follows we will take it to be constant. For the same reason we also ignore the dependence of conductivity on temperature which makes K a weak function of position as well. We then obtain for the distribution of temperature over the surface:

$$T^4 = T^4_{\text{pole}} \left(\sin^2 \Phi + \frac{K_{\perp}}{K_{\parallel}} \cos^2 \Phi \right) , \tag{14}$$

where T_{pole} is the temperature at the polar cap and $K_{\perp}/K_{\parallel}$ is expected to be small. Finally, for a purely dipolar field

$$\tan \Phi = 2 \cot \beta , \tag{15}$$

where β is the magnetic colatitude.

The temperature distribution of equations (14)–(15) is shown in Figure 2. In the case of isotropic conduction ($K_{\perp} = K_{\parallel}$) the temperature is constant over the star. As $K_{\perp}$ is reduced relative to $K_{\parallel}$, the equatorial regions grow cooler, reaching a temperature of absolute zero in this approximation for $K_{\perp}/K_{\parallel} = 0$. For physically plausible estimates of the ratio of perpendicular to parallel conductivities we expect a magnetic equator significantly cooler than the poles: note from Figure 2 that this regime is spread over a significant fraction of the surface area of the star. Finally, if polar cap heating is important, a further narrow step should be added at the poles.

III. OBSERVED PROPERTIES OF THE EMISSION

Itoh (1975), Brinkman (1980), and Cheng and Helfand (1983) have discussed the emissivity of neutron star surfaces. Best estimates indicate that both the total luminosity and spectrum should not deviate grossly from the blackbody, and in the following discussion we will assume a pure blackbody form.

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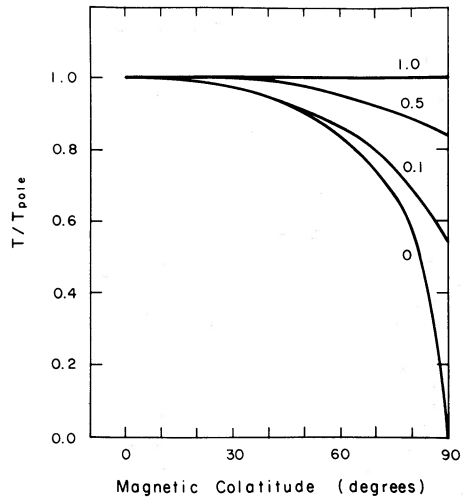

FIG. 2.—Distribution of surface temperature with magnetic colatitude, labeled by $K_{\perp}/K_{\parallel}$ (from eqs. [14]–[15])

The number of photons per second per steradian emitted by a blackbody of area dA in the energy interval $[E, E + dE]$ in the direction normal to dA is

$$\frac{2E^2 dE dA}{h^3 c^2[\exp(E/kT) - 1]} . \tag{16}$$

If the observer is located at an angle α from the normal to the surface element and is a distance D away, the number of photons per second arriving per unit detector area from dA in $[E, E + dE]$ is

$$\frac{2E^2 \cos\alpha \exp(-\tau) dE dA}{D^2 h^3 c^2[\exp E/kT) - 1]} , \tag{17}$$

where τ is the optical depth describing the attenuation of the signal by the interstellar medium. Finally, the count rate is equation (17) multiplied by the effective area of the detector, $\tilde{A}(E)$, and integrated over both the surface of the star and over energy:

$$\text{count rate} = \frac{2R^2}{D^2 h^3 c^2} \int_0^{\infty} dE \int_0^{2\pi} d\phi \int_0^{\pi} d\psi \frac{E^2 \tilde{A}(E) \exp[-\tau(E)] \sin\psi \cos\alpha}{\exp[E/kT] - 1} , \tag{18}$$

where R is the stellar radius and D the distance to the source. In performing these integrals we have used the interstellar absorption calculations of Brown and Gould (1970) and the IPC detector effective area data given by Giacconi *et al.* (1979).

a) Polar Cap

If equation (3) holds, emission from the polar cap dominates that from the rest of the star, and the problem simplifies considerably. The count rate from each pole becomes

$$\text{count rate} = \left[\frac{2}{h^3 c^2} \frac{A_{\text{cap}}}{D^2}\right] J(N_{\text{H}}, T) \cos\alpha(t) , \tag{19}$$

where

$$J(N_{\text{H}}, T) \equiv \int_0^{\infty} \frac{\tilde{A}(E) \exp[-\tau(E)] E^2}{\exp(E/kT) - 1} dE \tag{20}$$

and A_{cap} is the surface area of the polar cap. Values of the normalization integral $J(N_{\text{H}}, T)$ are given in Table 1.

The only time-dependent factor in equation (19) is $\cos\alpha$. Figure 3 illustrates the pulsar geometry: one of the magnetic poles is at geographic colatitude Ψ, and the line of sight to the observer is at colatitude θ. The angle between the normal to a magnetic pole and the line of sight to the observer is α. As the star rotates, α changes and the emission is modulated. We have

$$\cos\alpha(t) = \cos\theta \cos\Psi + \sin\theta \sin\Psi \cos\phi \tag{21}$$

(plus a similar term centered on the other pole), where ϕ is the pulsar phase.

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TABLE 1

THE FUNCTION $J(N_H, T)$

T (10^6 K)	N_H (cm^{-2}) 0	10^{21}	10^{22}
0.2	−5.24	−8.36	−15.09
0.6	−2.16	−4.07	−6.90
1.0	−1.37	−2.60	−4.19
1.4	−0.90	−1.63	−2.76
1.8	−0.52	−0.97	−1.85
2.2	−0.19	−0.50	−1.20
2.6	+0.10	−0.12	−0.71

NOTE.—Values of log $J(N_H, T)$ in units of keV^3 cm^2 (eq. [20]). We have used the IPC effective areas of Giacconi *et al.* 1979 and the interstellar attenuation results of Brown and Gould 1970.

This light curve undergoes a transition from double peaked to single peaked as the pulsar geometry changes. Presumably θ must be close to Ψ if the observer is to be illuminated by the radio beacon and detect the source as a pulsar. If $\theta = \Psi = 90°$ (an orthogonal rotator), each magnetic pole passes under the line of sight of the observer, and a double pulse is observed. In the opposite limit, $\theta = \Psi \ll 1$ (a nearly aligned rotator), only one pole is ever visible, the other remaining below the horizon as seen from the Earth. The transition occurs at $\theta = \Psi = 45°$. More generally, for $\theta \neq \Psi$, if $\theta + \Psi > 90°$, a double pulse is observed, and if $\theta + \Psi < 90°$, a single pulse is observed.

The light curve is very broad, with the full width at half-maximum given by

$$\mathrm{FWHM} = 2 \arccos \left[\tfrac{1}{2}(1 - \cot\theta \cot\Psi)\right] . \tag{22}$$

This relation is illustrated in Figure 4. The case of perpendicular orientations, $\theta = \Psi = 90°$, yields the narrowest pulse, for which the FWHM is 120°. As the magnetic pole is moved closer to the rotation axis, the pulse broadens. Also, the degree of modulation decreases: when θ drops below 30° (for $\theta = \Psi$), no solution to equation (22) exists since the intensity never drops to half its peak value.

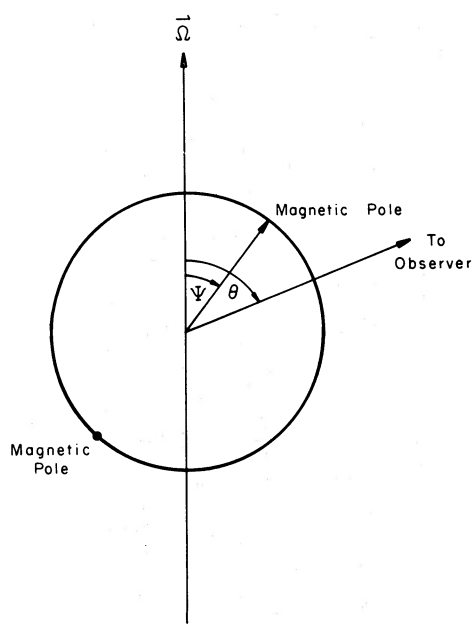

FIG. 3

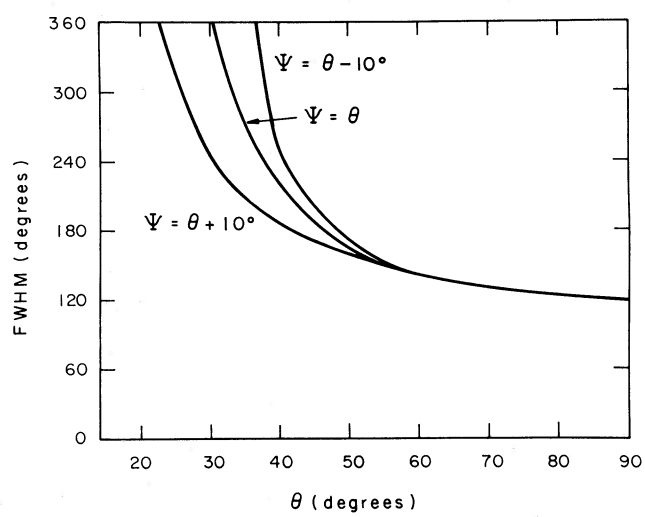

FIG. 4

FIG. 3.—Illustrating the pulsar geometry. One magnetic pole is at geographic colatitude Ψ, the line of sight to the observer at colatitude θ. In the configuration shown, the pulsar phase $\phi = \Omega t = 0$.

FIG. 4.—FWHM for the simple light curve (eq. [22]) of a small polar cap set on an effectively zero-temperature background

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The degree of modulation M of the emission, defined as peak intensity minus minimum intensity divided by minimum intensity, is

$$M = \tan\theta \tan\Psi \qquad \text{(double pulse)},$$

$$M = \frac{2}{\cot\theta \cot\Psi - 1} \qquad \text{(single pulse)}, \tag{23}$$

and is sketched in Figure 5. The more nearly aligned the rotator, the flatter the light curve and the smaller the modulation. However, as the colatitude Ψ of the magnetic pole is increased, it suffers greater and greater excursions from the observer's line of sight as the star rotates: the light curve grows more strongly modulated. The modulation grows arbitrarily high (implying zero emission between pulses) as Ψ approaches 45°: at this orientation, however, the second pole appears over the opposite horizon just as the first disappears from sight. At this geometry the source changes from single to double peaked, and M drops. Thereafter, as Ψ is increased to 90°, the modulation increases again.

b) *Anisotropic Heating from Below*

While analytically transparent, the above relations can only be used if equation (3) is satisfied. We now turn to a detailed integration of equation (18) in the opposite limit in which the temperature varies smoothly over the star. In the examples that follow we have assumed a temperature at the magnetic pole $T_{\rm pole} = 10^6$ K, a hydrogen column density to the source of $N_{\rm H} = 10^{21}$ cm^{-2}, and vanishingly small thermal conduction perpendicular to field lines.

In Figure 6 a typical set of "light" curves is illustrated for the geometry $\theta = \Psi$. The nearly aligned rotator shows a nearly flat light curve. As the magnetic pole is moved further south, the degree of modulation increases. The case $\Psi = 30°$ shows a significant pulselike character with a single peak. By $\Psi = 60°$ a weaker secondary peak has appeared, and finally the orthogonal rotator $\Psi = 90°$ shows two peaks of equal amplitude.

Figure 7 studies the effect of moving the observer away from a configuration in which one pole passes directly under the line of sight. The light curves differ only slightly from those of Figure 6: the effect is not dramatic. A qualitative feature is a somewhat flatter minimum than exhibited in Figure 6.

Figure 8 studies the degree of modulation of the source. The modulation is greatest for the orthogonal rotator. Also, it is greater for cooler stars than hotter ones. This effect arises because at low temperatures rotation can carry into view regions of the star so cool that the Planck function peaks well below the low-energy absorption edge of the interstellar medium. Observations sample only the high-energy tail, and the detected flux becomes highly sensitive to pulsar phase.

c) *Spectrum*

Although the emission mechanism is thermal in nature, the observed spectrum will not be blackbody. Observations sample different regions of the star which are at different temperatures. The sum of various Planck functions of different temperatures is not itself a Planck function.

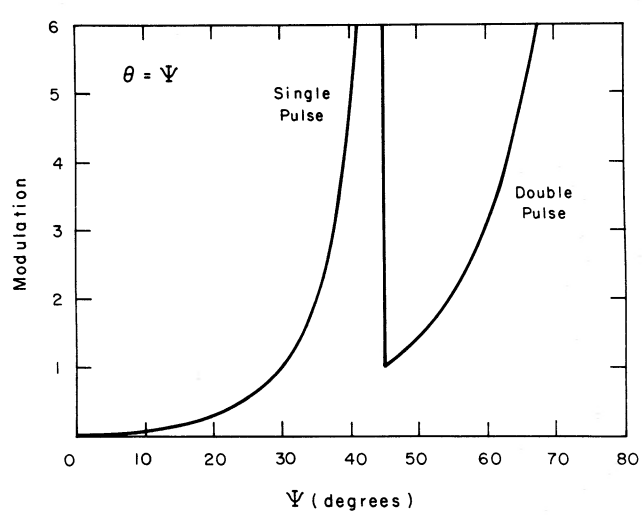

FIG. 5.—The degree of modulation (eq. [23]), defined as peak minus minimum divided by minimum, for the simple light curve of two magnetic poles set on an effectively zero-temperature background. We have assumed $\theta = \Psi$.

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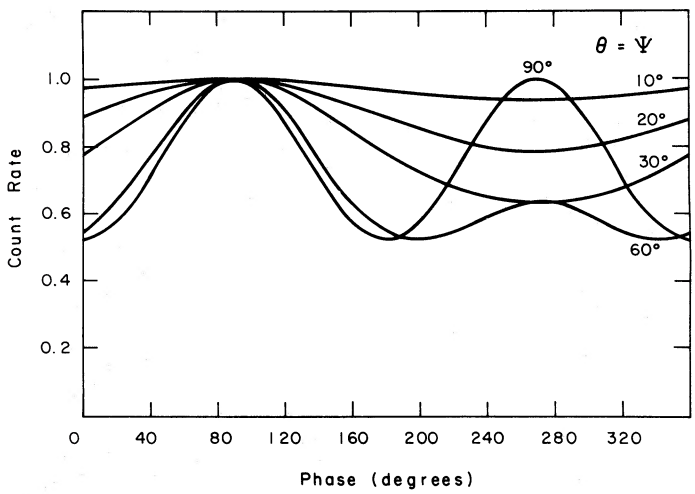

FIG. 6.—Counting rate in the *Einstein Observatory* IPC as a function of pulsar phase from equation (18). We have assumed $N_H = 10^{21}$ cm^{-2}, $\theta = \Psi$, $K_\perp/K_\parallel = 0$, and a polar temperature of 10^6 K. Curves are labeled by the colatitude Ψ of the magnetic pole.

The spectrum $I(E)$ is

$$I(E) = E^2 \tilde{A}(E) \exp[-\tau(E)] \int_0^p dt \int_0^{2\pi} d\phi \int_0^\pi d\psi \frac{\sin\psi \cos\alpha(t)}{\exp[E/kT] - 1}, \tag{24}$$

where the ϕ and ψ integrations cover the visible surface of the star and the t-integration covers rotation over one pulsar period. In a series of numerical experiments we have synthesized this spectrum and then fitted it to a pure blackbody function filtered through the ISM and IPC: the temperature of the best fitting function is what an observation would measure to be the temperature of the source. In all runs the hydrogen column was taken to be $N_H = 10^{21}$ cm^{-2}.

In one run the temperature distribution of equations (14)–(15) was adopted, a temperature at the pole $T_{pole} = 10^6$ K was assumed, and $K_\perp/K_\parallel$ was taken to be zero. The orthogonal rotator $\theta = \Psi = 90°$ was studied. The best-fit temperature was 0.93×10^6 K, and the post-fit residuals were less than 10% over the region in which the spectrum peaked: although the spectrum was not exactly blackbody, the deviations were not great, and the effect is probably unobservable. Similar behavior obtains at other temperatures. Adopting a nonzero conductivity across field lines smooths out the temperature distribution still more and results in still better fits to a pure blackbody.

Observations of the spectrum sensitive to pulsar phase were also modeled. The spectrum at the peak of the light curve was found to still more closely approach a Planck function. In a numerical experiment the above model was "frozen" at the point in the pulsar rotation at which the observer looks directly down upon the pole. The best-fit temperature was 0.95×10^6 K, and the residuals at most a few percent. (The same run also models a full rotation

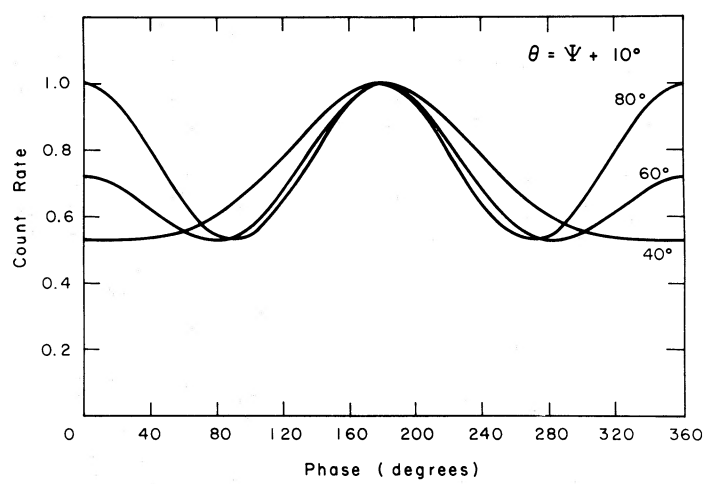

FIG. 7.—Same as Fig. 6, but with $\theta = \Psi + 10°$

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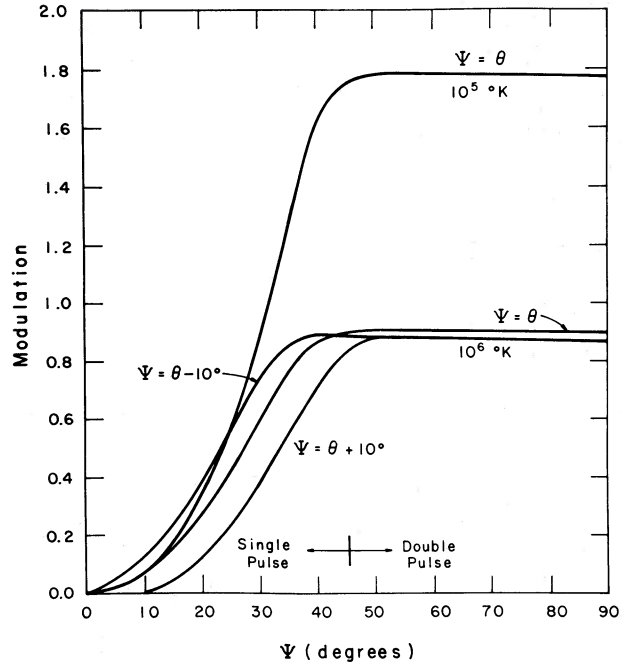

FIG. 8.—The degree of modulation of the light curve. We have assumed $N_H = 10^{21}$ cm^{-2} and $K_\perp/K_\parallel = 0$. Curves are labeled by the temperature of the magnetic pole.

of the aligned rotator $\theta = \Psi \ll 1$.) Observations at the minimum of the light curve show greater deviations from the blackbody, although still small: the edge-on configuration, in which the observer looks down upon the magnetic equator, yielded $T_{\rm fit} = 0.90 \times 10^6$ K.

The opposite limit, in which all emission arises from a small hot spot, yields a perfect blackbody spectrum. Only if the count rate from the hot spot just balances that from the remainder of the star does the spectrum depart significantly from the blackbody. By equation (3) this arises if the polar cap temperature is approximately equal to 7 times that of the rest of the stellar surface. If this condition is met, the observed spectrum is effectively the sum of two blackbodies at widely separated temperatures. In a numerical experiment the best-fit Planck function showed residuals of $\sim 10\%$ at the peak but $\sim 100\%$ slightly off-peak.

We conclude that, unless magnetospheric heating of the polar cap is fortuitously adjusted to balance that from the remainder of the star, the observed spectrum is blackbody in nature, and the best-fit temperature closely approximates that of the polar cap.

IV. TWO PULSARS

a) PSR 1055−52: The Curious Incident of the Dog in the Nighttime

In the accompanying paper (Cheng and Helfand 1983) the detection of X-rays from PSR 1055−52 is reported, and the possibility that the emission is blackbody discussed. Although a synchrotron spectrum best fits the data, a blackbody form cannot be ruled out at present. If the source is blackbody, the observed *absence* of modulation in this source has several important consequences.

Cheng and Helfand (1983) report an upper limit on the modulation M of 3%. This can only be understood if PSR 1055−52 is a nearly aligned rotator. In the context of the hot-spot model, equation (23) implies $\Psi \leq 7°$ if $\theta = \Psi$. Relaxing the restriction $\theta = \Psi$ softens this conclusion, but only slightly: if a maximum offset of 10° between the line of sight and the magnetic pole is allowed, Ψ can be increased to 14°. From Figure 8 a similar conclusion obtains within the context of the model of anisotropic heating from below. We have also explored the consequences of relaxing the assumption $K_\perp \ll K_\parallel$ and find that a 3% modulation implies $K_\perp/K_\parallel \approx 0.9$ (for $\theta = \Psi$). We regard such large values of $K_\perp$ to be physically impossible.

This result is difficult to understand if pulsar magnetic axes are randomly inclined relative to their rotation axis. If so, the probability that a pole lies within an angle Ψ of the spin axis is $1 - \cos \Psi$, which is 1% for $\Psi = 7°$ and 3% for $\Psi = 14°$. This cannot be ascribed to any alignment mechanism, for, with its period $P = 0.197$ s and spin-down age of 5×10^5 yr, PSR 1055−52 is relatively young. We note in this connection that Narayan and Vivekanand (1982*a*) have reached a somewhat similar conclusion in an analysis of the polarization swings of pulsars: they find a statistically significant preponderance of small values of θ which cannot be ascribed to alignment.

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The implication that PSR 1055−52 is nearly aligned is particularly significant in that this object shows an interpulse. When first discovered, interpulses were commonly regarded as emission from the magnetic poles opposite to those which generate the main pulses, although since then this view has changed somewhat (Manchester and Lyne 1977; Manchester 1978). If the X-ray emission from this object is indeed blackbody in nature, the case is closed in at least this particular instance.

A completely independent argument based on polarization data can also be made for the single-pole interpretation of the PSR 1055−52 interpulse. The position angle of polarization β for this object shows a linear swing of 1°.6 per degree of pulsar phase through the main pulse (McCulloch *et al.* 1978). Within the geometrical model of Radhakrishnan and Cooke (1969) β is given by

$$\tan \beta(t) = \frac{\sin \Psi \sin \phi(t)}{\sin \theta \cos \Psi - \cos \theta \sin \Psi \cos \phi(t)}, \tag{25}$$

where $\phi(t)$ is the pulsar phase (Manchester and Taylor 1977). The rate of change of β has a maximum at the center of the pulse given by

$$\frac{d\beta}{d\phi} = \frac{\sin \Psi}{\sin (\theta - \Psi)}. \tag{26}$$

These relations cannot be reconciled with the observed swing if an orthogonal rotator is assumed. If Ψ is set equal to 90° and $d\beta/d\phi$ fixed at 1.6, equation (26) implies $\theta - \Psi = 39°$, in conflict with current notions that the width of the pulsar beacon is no more than $\sim 10°$. Fixing $d\beta/d\phi$ at its observed value and decreasing Ψ, equation (26) yields progressively closer cuts of the line of sight across the pole: only for $\Psi < 15°$ does $\theta - \Psi$ lie below 10°. Detailed plots of equation (25) also show a strong S-shape for nearly orthogonal rotators, and large excursions in β from beginning to end of a pulse. The assumption of near alignment, however, shows more nearly linear swings with smaller $\Delta\beta$.

If interpulses arose from opposite magnetic poles than main pulses, they would be expected to lie $\Delta\phi = 180°$ in phase away from main pulses, at least in an average sense. In PSR 1055−52 the offset is 150°. Among the six pulsars showing interpulses listed by Manchester and Taylor (1977) only two show $\Delta\phi = 180°$; the average is 164°.

The strongest arguments against the single-pole interpretation of PSR 1055−52 are the frequency independence of the separation between its main and interpulses (Rankin 1982) and the low *a priori* probability of its alignment. We note that PSR 0950+08 also shows an interpulse whose separation $\Delta\phi$ is frequency independent, although the correlations between main and interpulses in this source are difficult to understand on the basis of the two-pole interpretation (Hankins and Cordes 1981). Finally, Fowler and Wright (1982) have shown that the PSR 1822−09 interpulse participates in the intensity fluctuations of the main pulse and in the mode changes exhibited by this source, once again implying a tight communication between the two. This object also shows a $\Delta\phi$ independent of observing frequency. A possible model to account for the frequency independence of $\Delta\phi$ in the single-pole model has been given by Narayan and Vivekanand (1982*b*).

b) *PSR 1509−58*

PSR 1509−58 is unique in having been discovered as an X-ray pulsar before it was detected as a radio pulsar (Seward and Harnden 1982; Seward *et al.* 1983; Manchester, Tuohy, and D'Amico 1982). Its period $P = 0.15$ s is relatively large, and its period derivative $\dot{P} = 1.5 \times 10^{-12}$ the greatest known of any pulsar.

Helfand (1982) has emphasized the remarkable character of the X-ray pulse shape exhibited by this object. It is very broad, with FWHM $\approx 25\%$ of the pulse period, in contrast to $\sim 9\%$ for its radio pulse and an average of $\sim 3\%$ for the radio pulsars as a class. It is also significantly broader than the X-ray pulse of the Crab pulsar. Here we explore the possibility that the X-ray emission from this source is blackbody in nature.

We first consider emission from a small polar cap. The light curve of Seward and Harnden (1982) exhibits a single pulse, which constrains the geometry to $\theta + \Psi < 90°$. The modulation, best estimated from the HRI data which are relatively weakly contaminated by the extended source, is $M \gtrsim 5$. By equation (23) this places a second constraint on the geometry. In Figure 9 these two constraints are graphed. Requiring $|\theta - \Psi| < 15°$ to insure that the source be detected as a radio pulsar limits θ and Ψ to lie within the crosshatched region. The configuration is close to $\theta = \Psi = 40°$.

The IPC count rate reported by Seward and Harnden (1982) is 0.3 counts s^{-1}, and the hydrogen column density is 1×10^{22} cm^{-2}. From equations (1) and (19)–(21),

$$\langle \text{count rate} \rangle = 7.3 \times 10^{-4} \left[\frac{R}{10 \text{ km}}\right]^3 J(N_{\rm H}, T) \langle \cos \alpha \rangle , \tag{27}$$

where, for $\theta + \Psi < 90°$,

$$\langle \cos \alpha \rangle = \cos \theta \cos \Psi . \tag{28}$$

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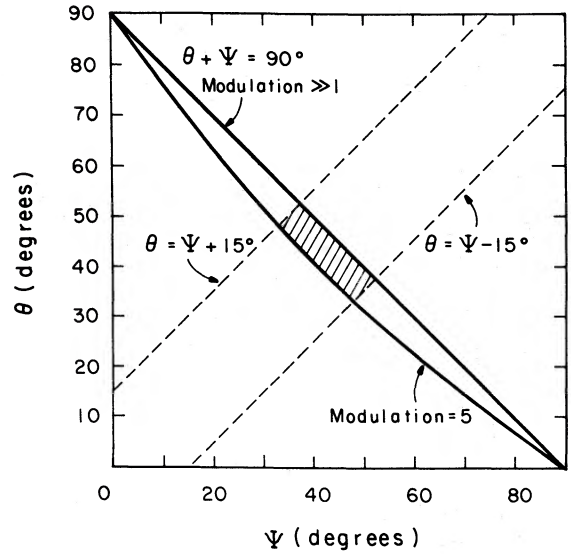

FIG. 9.—Illustrating the constraints placed on the geometry of PSR 1509−58 by the observed light curve, θ and Ψ must lie within the crosshatched region.

Specializing to $\theta + \Psi = 40°$, $N_H = 10^{22}$ cm^{-2}, and a stellar radius of 10 km yields a temperature at the magnetic pole of 2×10^7 K. The observed count rate can be obtained at a lower temperature by going to larger (i.e., less massive) stars; the choice $R = 20$ km yields $T_{pole} = 9 \times 10^6$ K. Alternatively, the area of the emitting surface might be significantly larger than the region defined by the open field lines of a perfect dipole. We note that this appears to be the case for PSR 1055−52: Cheng and Helfand (1983) estimate the linear dimension of their source, if blackbody, to be ~ 7 times the R_{cap} given by equation (1). Adopting this choice for PSR 1509−58 as well yields $T_{pole} = 6 \times 10^6$ K.

The light curve predicted by the model is compared with the observed pulse shape in Figure 10. The fit is generally good, although there is some indication that the predicted pulse is somewhat too broad. This may imply a beaming mechanism is at work, although the disparity is not great.

No acceptable model can be found if emission from the magnetic poles does not dominate that from the remainder of the star. From runs similar to those shown in Figure 8, the observed degree of modulation forces one to low

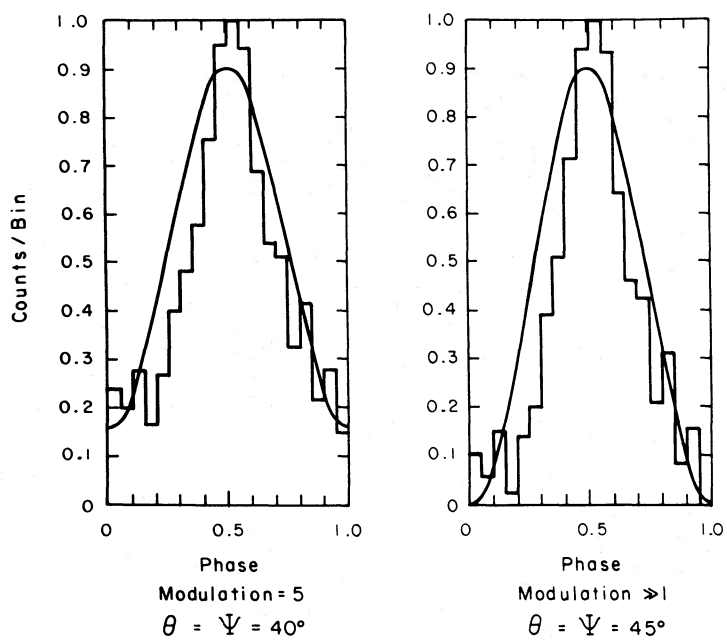

FIG. 10.—Comparing the observed and predicted light curves of PSR 1509−58

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temperatures. Constraining $\theta \approx \Psi < 45°$ to obtain a single pulse implies $T_{\rm pole} < 10^5$ K. However, at such temperatures the predicted IPC count rate is orders of magnitude smaller than observed.

We conclude that a model for PSR 1509−58 can be developed in which the X-ray component of the emission is blackbody in nature. Within this model magnetospheric heating of the polar regions is very strong, and there is some indication that it extends over a significantly larger area than a classical polar cap. We wish to point out that it is somewhat surprising that a simple model based on a purely dipolar configuration succeeds at all. The dynamical implosion in which a neutron star is formed is turbulent in nature, and the stellar field, whatever its initial form, is scrambled in the collapse. Upon formation, the pulsar magnetic field could not remotely approach a dipole configuration. Observations of X-ray light curves map the *surface* field, as opposed to inferences drawn from behavior far from the star (e.g., at the speed-of-light cylinder) at which the higher moments have dropped off. The observation of PSR 1509−58 is unique in this regard, and it may imply that magnetic field decay has acted to smooth out all but the dipole component.

It may be significant that this source has the highest period derivative of any known pulsar. Furthermore, PSR 0655+64, the pulsar with the smallest known period derivative, has one of the lowest limits on its X-ray luminosity (Helfand, unpublished). The implication is that the stronger the torque, the more intense the backflow of energy from the magnetosphere onto the star, and the stronger the X-ray emission from the polar cap.

Finally, we note that if our interpretation is correct, Seward and Harnden (1982) have identified the smallest known object at stellar distances. The radiating surface is one-half mile in extent and is ~14,000 lt-yr away.

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